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Abstract. TEXT

1 Introduction

2 Data and Methods

2.1 Site and database

An extensive dataset ranging the years 2018 to 2023 was used. In particular, Cloudnet **?** post-processed high level data and radiometer humidity and temperature profiles. These products were inferred by measurements carried out with a set of ground-based remote sensing instruments - managed at the ACTRIS-CCRES Granada site facility (CCRES stands for Centre for Cloud Remote Sensing) at 37°2E, 3.6°S and 680 m, Southeast Spain. This station is as part of the University of Granada (UGR) and it is integrated into the AGORA observatory. Figure 1 shows the station, located within a broad intramontane depression in the foothills of the tallest mountain massif of the Iberian Peninsula, Sierra Nevada.

The Cloudnet processed data was determined by a synergy between a RPG-HATPRO radiometer, a CHM15k ceilometer, a RPG-FMCW 94 GHz cloud Doppler radar and the European model ECMWF (an acronym for European Centre for Medium-Range Weather Forecasts). These instruments have been continuously operated since 2018, and their data has been processed by the CloudnetPy **?**, performing reflectivity attenuation correction, error estimation and quality flag assignment. Particularly, attenuation corrections assists in prevention misclassification of atmospheric elements through liquid water content (LWC) estimation. This variable is obtained by cloud boundaries provided by lidar and radar synergy, and radiometer liquid water path (LWP) **?**. As a result, uncertainties retrieving LWP during a raining event can be high enough to frustrate attenuation corrections. Consequently, rain events were exclude from cloud analysis in the section **??**, reducing misclassification occurence.

Processed data were sampled in a common grid with time resoltion of 30 s, and height bins according to cloud radar range resolution, which differ according to radar chirp table configuration. Figure 2 displays changes in the radar vertical bin size across dataset time serie, and table 1 associate bin sizes with their respective height intervals. This table also gives operational

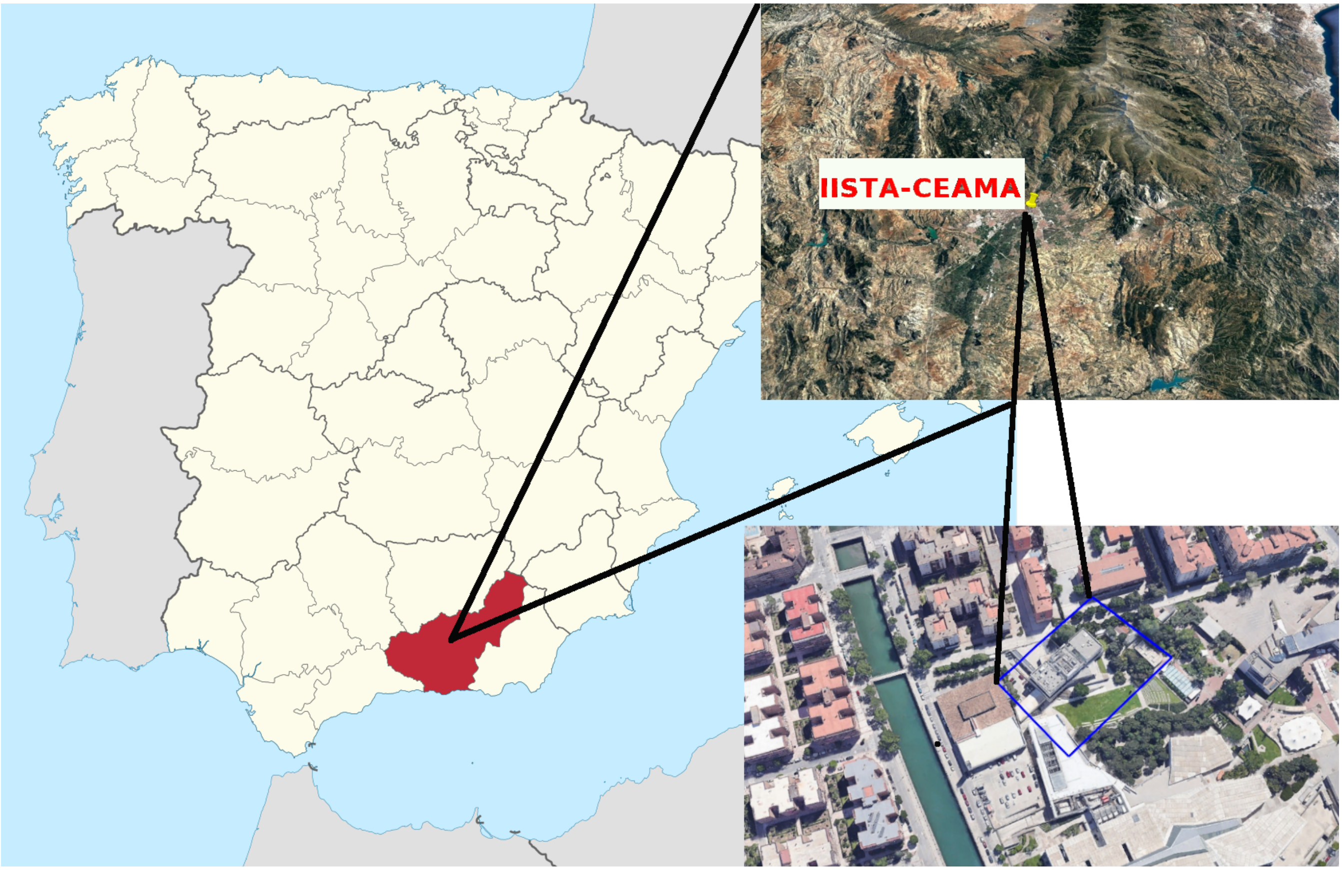

Figure 1. Geographic location of the ACTRIS CCRES Granada station. Red area is the province of granada, while blue poligon indicates the station facility, the Andalusian Institute for Earth System Research. Image adapted from wikipedia with Google Earth software.

information of instruments at Granada site and their main post-processed variables. Differently, microwave radiometer humidity and temperature profiles were not processed by CloudnetPy, but were provided by its own manufacturing software. High level data included in the analysis were target classification, liquid water path (LWP) and ice water path (IWP), where target classification apportioned to classify cloud layers as explained in section **??**. The classification method was adapted from **?** and **?**.

Typical variables of each instruments in table 1 in Granada station is shown in Figure **??**. The corresponding instrument is depicted on the right side.

2.2 Cloudnet particle classification scheme

Target classification product assigns an integer ranging from 0 to 10 at each pixel in the Cloudnet processed grid. Each number are related to atmospheric elements in the following order: clear sky, droplets, drizzle or rain, drizzle & droplets, ice, ice

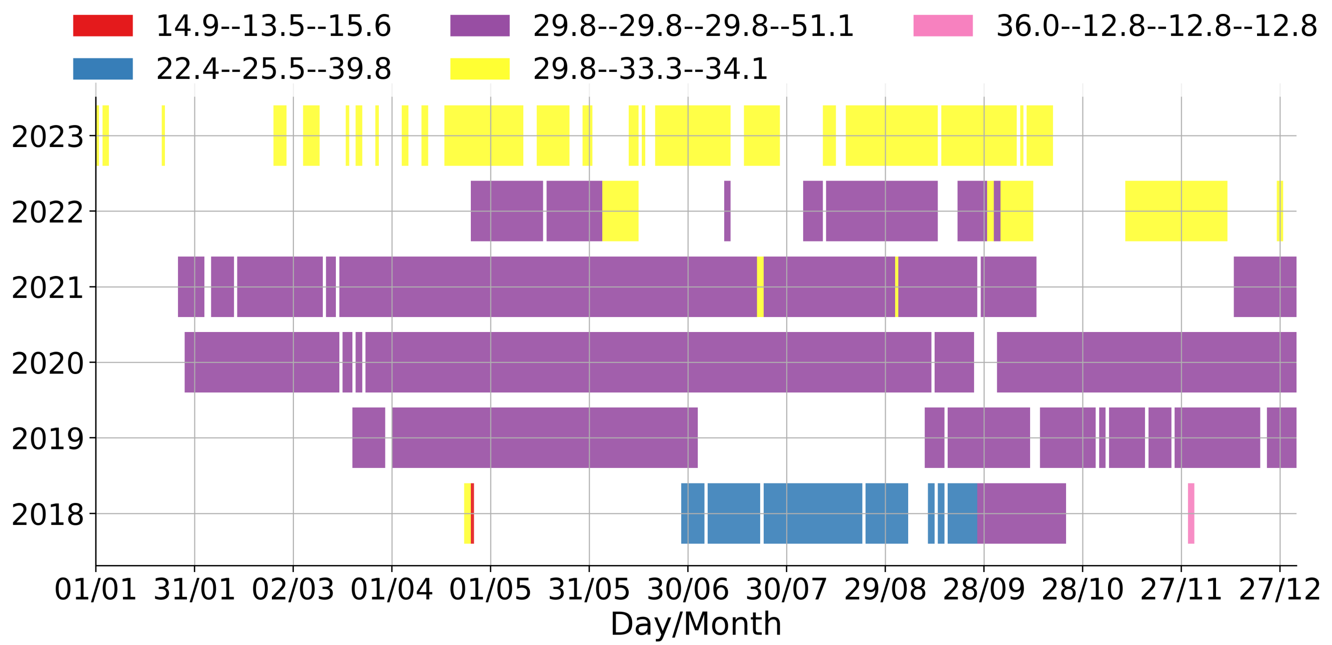

Figure 2. Overview of cloud radar vertical bin sizes depending chirp table configuration.

Table 1. This is a sample table caption and table layout.

Instrument	**Height resolution (heigh interval) (m)**	**Integration time (s)**	**Variables**
FMCW 94 GHz Cloud Doppler Radar	14.9 (100-1200), 13.5 (1200-4400), 15.5 (4400-10000)	1.10	Mean Doppler vellocity (m/s)
	22.4 (100-924), 25.6 (924-8455), 39.7 (8455-12000)	3.2	Reflectivity (dBz)
	29.8 (100-1200), 29.8 (1200-4900), 29.8 (4900-8500), 51.1 (8500-13500)	3.57	Linear depolarisation ratio (dB)
	29.8 (100-1200), 33.3 (1200-5500), 34.1 (5500-11000)	3.26	Spectral width (m/s)
	36 (180-1476), 12.8 (1476-3998), 12.8 (3998-8158), 12.8 (8158-1200)	2.81	
MWR RPG HATPRO	200 (0-2000), 400 (2000-5000), 800 (5000-10000)	10	Humidity profiles (g m^{-3})
	50 (0-1200), 200 (1200-5000), 400 (5000-10000)	10	Temperature profiles (K)
	–	2	LWP, IWP (kg m^{-2})
CHM15k Nimbus Ceilometer (1064 nm)	15	15	Attenuated backscatter coefficient (sr^{-1} m^{-1})

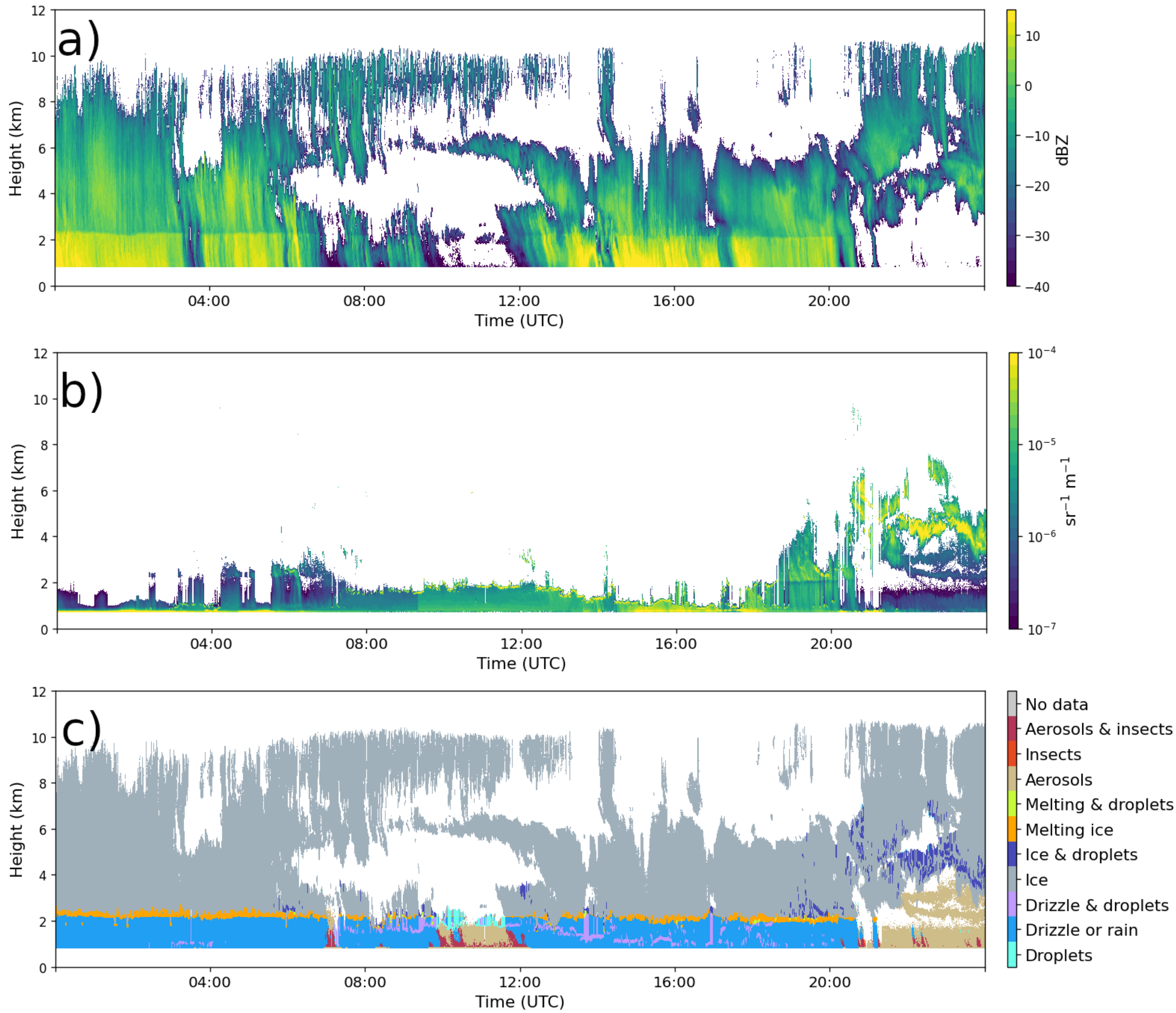

Figure 3. Should change figure - Reflectivity, Att. backscatter coeff and LWP

& droplets, melting ice, melting & droplets, aerosol, insects, aerosol & insect. These targets were determined through radar and lidar variables in combination with thermodynamic profiles from the ECMWF model. In particular, wet bulb temperature (T_w) profiles is used to establish warm and cold regions to classify liquid and ice hydrometeors. For instance, ice and melting particles can only exist in cold regions, in other words, when T_w < 0. Further, liquid droplets boundaries were resolved with lidar attenuated backscatter coefficient (β), and radar reflectivity. The first is sensible to small liquid droplets present in cloud base, being used to identify lower limits of droplet layers and also upper limits when there is no radar reflectivity above the heigh lidar signal was totally attenuated. In addition, liquid layers respect a threshold of T_w < -40 °C, since it is not possible to keep water in liquid phase bellow that temperature. Hence, liquid droplets assigned as falling were classified as drizzle or rain droplets, where falling is attributed through radar echos analysis once particles boundary were already determined. Therefore, falling cold particle were just classified as ice, without distinguishing ice from precipitating ice. Next, melting layers were identified at the height of maximum wet bulb temperature divergence. Pixels with divergence surpassing 0.0075 s^{-1} within 150 m of height deviation were also classified as "melting". Furthermore, more than one class of particle can be settled in the same pixel, allowing the identification of mixed phase hydrometeors, like ice coexisting with supercooled liquid droplets and melting ice particles coexisting with cloud liquid droplets. A more detailed description of particle apportionment of Cloudnet algorithm can be found on **?**.

2.3 Assortment algorithm

The assortment algoritm classifies groups of hydrometeors and cloud layers. In terms of hydrometeors, this algorithm generate masks of 3 groups: liquid, ice and "total", where liquid class encompass all possible liquid pixels, which are "droplets", "drizzle or rain" and "drizzle & droplets". Ice encompass only "ice" pixels, and total is all possible hydrometeors, whic are "droplets", "drizzle or rain", "drizzle & droplets", "ice", "ice & droplets", "melting ice" and "melting & droplets".

Subsequently, cloud layers in atmospheric column were classified trought pixel sequence inquiry. These pixels sequences were devided in the following 5 groups according to **?**: liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only "ice" pixel sequence was encountered. As well as, mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified wheter at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Diferent from previous studies, a threshold of 20 bins (256-1022 m) bellow cloud was set to reject precipitation events, assuming that rain droplets are not falling from the actual cloud layer. In all cases, for being considered a cloud layer, a minimun sequence of 3 cloud pixels (38-153 m) (as (**?**)) must be satisfied, in addition to a minimun sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Arosol & insect" to separate multilayers. Therefore, free hydrometeor layers in the neghborhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on **?**, but theses values are not currently well defined, their determination has been guided by empirical experience.

3 Thermodinamic conditions

3.1 HEADING

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4 Conclusions

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Code availability. TEXT

Data availability. TEXT

Code and data availability. TEXT

Sample availability. TEXT

Video supplement. TEXT

Appendix A

A1

Author contributions. TEXT

Competing interests. TEXT

Disclaimer. TEXT

Acknowledgements. TEXT

References

REFERENCE 1

REFERENCE 2